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# 目次

|                                                                                              |    |
|----------------------------------------------------------------------------------------------|----|
| 第1章 英訳                                                                                       | 1  |
| 1.1 Introduction . . . . .                                                                   | 1  |
| 1.2 Related Work . . . . .                                                                   | 2  |
| 1.3 Human–Robot Interaction with Correction and Suggestion (HRI-CAS) . .                     | 4  |
| 1.3.1 Implementation Platform and System . . . . .                                           | 5  |
| 1.3.2 MR System Implementation . . . . .                                                     | 6  |
| 1.4 Module of Human–Robot Interaction with Correction and Suggestion (HRI-<br>CAS) . . . . . | 7  |
| 1.4.1 Target Bottle Inference (TBI) Module . . . . .                                         | 7  |
| 1.4.2 Base Relocation based on Inverse Reachability Map (BRI) . . . . .                      | 10 |
| 1.4.3 Human-Intent Adaptive Grasp Manipulation (HIGM) . . . . .                              | 14 |
| 1.4.4 Significance of Module Integration . . . . .                                           | 17 |
| 1.5 Experiment on BRI . . . . .                                                              | 18 |
| 1.5.1 Experimental Setup . . . . .                                                           | 18 |
| 1.5.2 Experimental Results . . . . .                                                         | 19 |
| 1.6 Experiment2 . . . . .                                                                    | 19 |
| 1.6.1 Experimental Setup . . . . .                                                           | 19 |
| 1.6.2 Experimental Results . . . . .                                                         | 22 |
| 1.7 Conclusions . . . . .                                                                    | 22 |
| 参考文献                                                                                         | 23 |

# 图 目 次

|      |                                                                  |    |
|------|------------------------------------------------------------------|----|
| 1.1  | Figure1 . . . . .                                                | 5  |
| 1.2  | Figure2 . . . . .                                                | 5  |
| 1.3  | Figure3 . . . . .                                                | 7  |
| 1.4  | candidate bottle is classified by the Reachability Map . . . . . | 9  |
| 1.5  | Inferred target bottle . . . . .                                 | 9  |
| 1.6  | RM/IRM . . . . .                                                 | 10 |
| 1.7  | Figure7 . . . . .                                                | 11 |
| 1.8  | Figure4 . . . . .                                                | 12 |
| 1.9  | Figure4 . . . . .                                                | 13 |
| 1.10 | Figure10 . . . . .                                               | 15 |
| 1.11 | Figure11 . . . . .                                               | 16 |
| 1.12 | Figure12 . . . . .                                               | 17 |
| 1.13 | Overview of BRI pipeline . . . . .                               | 18 |
| 1.14 | Sequential pick-and-place before/after . . . . .                 | 20 |
| 1.15 | Sequential pick-and-place . . . . .                              | 20 |
| 1.16 | Sequential pick-and-place . . . . .                              | 21 |

# 第1章 英訳

## 1.1 Introduction

In recent years, Human-Robot Collaboration (HRC) has attracted significant attention as an important research domain in which humans and robots cooperate to accomplish tasks in diverse working environments[1][2]. The aim is not only for robots to interpret human intentions more accurately but also to enable smooth and intuitive communication. When working in shared spaces, the synergy between humans and robots can be expected to improve both efficiency and safety.

One of the major challenges in HRC lies in the discrepancy between human intention and robotic perception. Such mismatches often cause reductions in task efficiency and increase the difficulty of operation. In particular, operational mismatches stemming from the differences in embodiment between humans and robots represent a fundamental bottleneck in collaborative work. To address this issue, approaches that leverage Mixed Reality (MR) to intuitively map human operations onto robots have been actively studied. MR enables users to recognize robotic motion as an extension of their own body, thereby fostering more natural and seamless collaboration [3][4][5][6][7].

Frameworks for intuitive operation support in HRC have also been reported not only with MR but also through Augmented Reality (AR) and Virtual Reality (VR). For example, VR-based methods have been proposed for imitation learning and motion data collection, allowing robots to efficiently acquire manipulation skills [8] [9][10]. Meanwhile, AR-based approaches have sought to present robot states and task procedures to the user, thereby enhancing task comprehension [11] [12][13]. Although these works have shown the effectiveness of bridging recognition gaps between humans and robots, limitations remain in terms of immersion and interaction with the real-world environment. In this context, MR is particularly promising because it allows for the seamless integration of real-world environments with virtual information, thus enabling more intuitive and adaptive HRC.

A representative prior study is the work by Esaki et al., entitled Immersive Robot Teleoperation Based on User Gestures in Mixed Reality Space (IRT-MRO)[14]. IRT-MRO demonstrated the effectiveness of HRC using MR by enabling humans to intuitively manipulate virtual objects in MR space, with the robot recognizing and executing corresponding positional and postural changes. However, several challenges remain unresolved, including the integration of environmental perception, the assurance of stable robot posture, and the autonomy of base relocation strategies.

To address these challenges, this study proposes Human-Robot Interaction with Correction and Suggestion (HRI-CAS). The system integrates three key components: Target

Bottle Inference (TBI), which estimates the target object based on user operation features; Base Relocation based on the Inverse Reachability Map (BRI), which provides a base movement strategy utilizing the IRM; and Human-Intent Adaptive Grasp Manipulation (HIGM), a grasping framework that compensates for the embodiment gap between humans and robots. By combining these components, HRI-CAS enables stable collaborative work independent of the user's skill level and aims to realize bidirectional and interactive HRC through MR space.

## Contributions

The main contributions of this study can be summarized as follows:

- **Proposal of the Target Bottle Inference (TBI) module** By integrating user operation features such as hand proximity, contact, gaze, and arm motion, and weighting them according to reachability, the TBI module infers the intended object of manipulation. This approach suppresses unnecessary information presentation while supporting intuitive target selection.
- **Design and realization of the Base Relocation module based on the Inverse Reachability Map (BRI)** By utilizing the Inverse Reachability Map[17], the system computes feasible base positions for grasping and visualizes candidate paths in MR space. Users can edit these paths interactively, thereby achieving a balance between autonomy and interactivity in determining whether base relocation is necessary.
- **Development of the Human-Intent Adaptive Grasp Manipulation (HIGM) framework** By combining pregrasp generation, nullspace optimization, and a finite state machine (FSM), the system continuously harmonizes user operations with robot manipulability. This enables seamless collaborative manipulation based on mutual correction and suggestion.

Through these contributions, the proposed system allows robots to respect user intentions while providing appropriate correction and suggestions, ensuring stable task execution regardless of user expertise. Furthermore, the proposed HRI-CAS framework demonstrates applicability to real-world tasks where humans and robots closely collaborate, such as in manufacturing sites, remote maintenance, and medical support. This work is thus expected to contribute to the further development of HRC research.

## 1.2 Related Work

Robots are expected to support human life and activities in a wide range of fields such as medicine, manufacturing, space exploration, and agriculture. However, many challenges remain for humans and robots to work collaboratively in the same space. One of these is the lack of mutual understanding of intentions and actions. Humans often

find it difficult to accurately predict the capabilities and behaviors of robots, while robots cannot fully interpret humans' high-level goals. As a result, "gaps of execution" and "gaps of evaluation" arise, hindering smooth collaboration. Conventional interfaces have mainly relied on joysticks or keyboards, but visualization on 2D displays provides insufficient depth information, limiting the operator's spatial awareness and sense of immersion.

To address these challenges, research on Human-Robot Collaboration (HRC) using Mixed Reality (MR) has made significant progress in recent years. By overlaying information about robot states and environmental conditions in MR space, users can perform more intuitive operations, and improvements in efficiency for teleoperation and collaborative tasks can be expected [3] [4].

In addition, many studies have explored applications of AR and VR. In VR, methods have been proposed to support skill acquisition by robots through imitation learning and motion data collection [8] [9], while in AR, attempts have been made to assist workers' understanding by presenting task procedures and robot states [11] [12]. Nevertheless, these approaches still face limitations in interaction with the real environment and in providing a sense of immersion.

In this context, MR, which can integrate the real world with virtual information, is receiving particular attention as a promising foundation technology for HRC. Previous MR-based HRC research can largely be classified into two directions.

First, research focusing on **visualization of robot states**. This direction projects sensor data and robot internal states into MR space, thereby supporting users' situational awareness and decision-making [15]. While this approach contributes to enhancing environmental understanding and reducing cognitive load, direct intervention in robot motion planning and control remains limited.

Second, research focusing on **intuitive mapping of user operations**. By converting hand tracking and gesture inputs into robot motions, attempts have been made to enhance operability and immersion [3][4][5][6][7]. Although this approach allows the operator's intentions to be reflected more naturally, the stability of the operation and the guarantee of robot manipulability are often insufficient.

In this category, the work of Esaki et al. is representative. They proposed the **Immersive Robot Teleoperation Based on User Gestures in Mixed Reality Space (IRT-MRO)** [14]. In IRT-MRO, objects are placed in MR space, and users directly manipulate their positions and orientations. The manipulated results are transferred to the robot side, enabling the robot to execute the same operations in the physical world. This demonstrated the effectiveness of MR in improving the intuitiveness of robot teleoperation.

However, IRT-MRO also leaves several issues unresolved. First, because the system simply transfers the object manipulation performed by the user in MR space, it does not take into account whether the robot can actually reach the target object or grasp it

stably. Second, the posture of the robot arm is not always guaranteed to remain safe, and problems such as singularity or joint limit violations can occur. Third, when an object is located outside the reachable area of the robot, the system does not provide an appropriate relocation strategy for the robot base, leaving the operation incomplete.

In summary, while existing studies have clearly demonstrated the effectiveness of MR for HRC, they have not yet resolved several key challenges, such as the lack of integration of environmental information, the absence of stable operation considering redundant degrees of freedom, and the insufficient autonomy of base relocation strategies. Moreover, inherent limitations of MR technology itself remain, including the lack of haptics and constraints on real-time processing performance [16]. Against this background, this study proposes Human–Robot Interaction with Correction and Suggestion (HRI-CAS).

### 1.3 Human–Robot Interaction with Correction and Suggestion (HRI-CAS)

**Human–Robot Interaction with Correction and Suggestion (HRI-CAS)** addresses several limitations observed in conventional MR-based HRC systems, including insufficient utilization of environmental information, lack of stable posture guarantees, and inadequate autonomy in base relocation strategies.

This system is built upon the foundation of *Immersive Robot Teleoperation Based on User Gestures in Mixed Reality Space (IRT-MRO)* and improves upon the issues that remained in IRT-MRO. Whereas IRT-MRO employed a Unity-based holographic application on the HoloLens 2, the proposed system adopts a Unity-based passthrough application using the Meta Quest. Although both devices are capable of projecting real-world objects into MR space, they differ significantly in their display methods.

The HoloLens 2 adopts an optical see-through display, allowing the user to perceive both the real environment and virtual objects through a transparent screen. However, its limited field of view and restricted brightness have been widely noted. In contrast, as illustrated in Figure 1.1, the Meta Quest employs a passthrough display system, which projects images captured by external cameras onto the display, enabling a wide field of view and high-resolution rendering. This provides the user with a more natural and immersive MR experience, enhancing both environmental understanding and intuitive operation.



Fig.1.1: Figure1

### 1.3.1 Implementation Platform and System

In this study, the experimental platform adopted was the **KUKA YouBot**. The robot is a mobile manipulator equipped with an omnidirectional base, a 5-DOF manipulator, and a two-finger gripper. Within the proposed system, changes in the position and orientation of target objects detected in MR space, as well as user manipulations, are reflected directly in the robot's motion. This ensures smooth and seamless interaction between the human and the robot. Furthermore, the robot's behavior is immediately fed back into MR space, providing the operator with precise visual information in real time.

The graphical architecture of the system is illustrated in Figure?? and is composed of the following three modules:

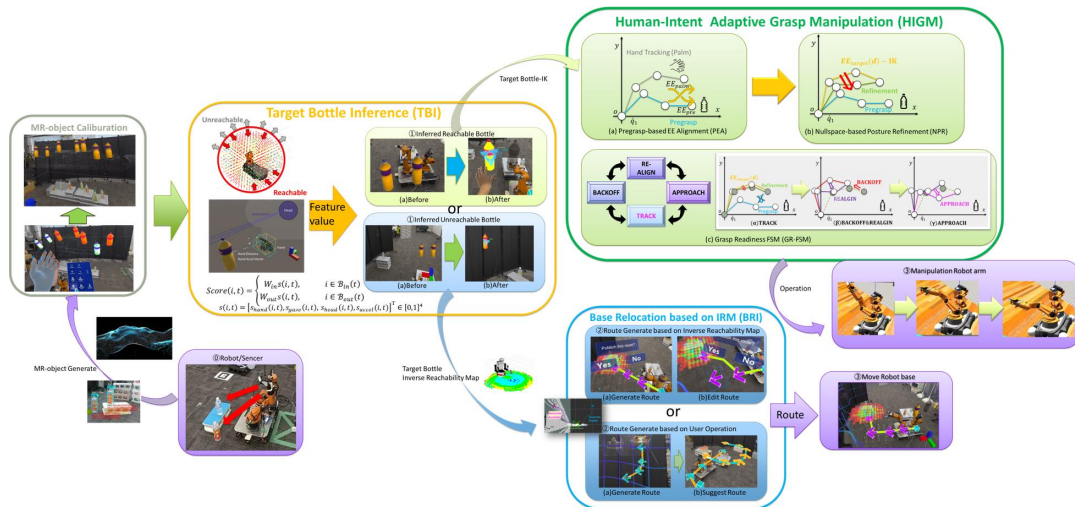


Fig.1.2: Figure2



**Target Bottle Inference (TBI) module:** This module tracks the user ’ s gaze, head orientation, and hand motion within MR space, extracting geometric relationships with candidate bottles as features. These multimodal features are then processed through an integrated scoring framework to infer the bottle that the user intends to manipulate. By jointly evaluating diverse behavioral cues, the TBI module enables highly accurate and real-time target selection.

**Base Relocation based on Inverse Reachability Map (BRI) module:** When the target bottle lies outside the robot ’ s reachable workspace, this module becomes active. First, an Inverse Reachability Map (IRM) is generated based on the Reachability Map (RM)[17], and candidate base positions that make the target reachable are extracted. Next, an optimization problem is solved considering both the IRM score and the movement cost, yielding the optimal base configuration. Additionally, candidate paths are visualized within MR space, allowing users to refine or adjust the proposed routes interactively.

**Human-Intent Adaptive Grasp Manipulation (HIGM) module:** This module aims to reconcile the differences between the user ’ s intended manipulation and the robot ’ s physical and kinematic constraints. Specifically, it integrates reference postures obtained from hand-tracking information with robot-preferred pregrasp postures, and applies nullspace optimization to adjust joint configurations. As a result, the user can achieve stable grasping with natural operational feedback, without needing to be aware of the robot ’ s inherent kinematic constraints.

Through the integration of these modules, the proposed system achieves adaptive and intuitive motion planning that was not sufficiently addressed in IRT-MRO. The robot is now able to select actions according to the user ’ s intended target, while providing adaptive grasp correction without requiring the user to account for robot-specific constraints. This significantly reduces dependency on the user ’ s operational expertise and ensures smooth, stable operation even for novice users.

### 1.3.2 MR System Implementation

The MR system is implemented on **Unity** and **MRTK3 (Mixed Reality Toolkit)**, in compliance with the OpenXR standard. The user can manipulate MR objects directly through hand gestures, offering an intuitive and natural input interface. Moreover, by presenting the robot ’ s behavioral intentions through MR objects, the system improves the operator ’ s understanding of the robot ’ s decision-making process, thereby enhancing human-robot collaboration.

## 1.4 Module of Human–Robot Interaction with Correction and Suggestion (HRI-CAS)

### 1.4.1 Target Bottle Inference (TBI) Module

**Definition of Features** : The TBI module aims to estimate the user’s intended target object based on bodily movements. Figure1.3 illustrates the distribution of operational features in the information space.

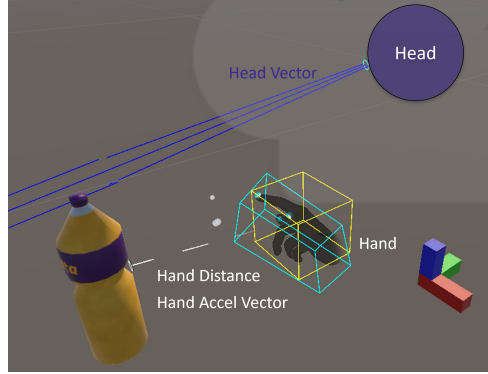


Fig.1.3: Figure3

Let the position of a target object  $o_i$  be denoted as  $\mathbf{p}_i(t)$ , and the positions of the user’s eyes, head, and hand be  $\mathbf{p}_{eye}(t)$ ,  $\mathbf{p}_{head}(t)$ , and  $\mathbf{p}_{hand}(t)$ , respectively. For each object, the normalized direction vectors from these body parts are defined as:

$$\mathbf{d}_i^{eye}(t) = \frac{\mathbf{p}_i(t) - \mathbf{p}_{eye}(t)}{\|\mathbf{p}_i(t) - \mathbf{p}_{eye}(t)\|}, \quad \mathbf{d}_i^{head}(t) = \frac{\mathbf{p}_i(t) - \mathbf{p}_{head}(t)}{\|\mathbf{p}_i(t) - \mathbf{p}_{head}(t)\|}, \quad \mathbf{d}_i^{hand}(t) = \frac{\mathbf{p}_i(t) - \mathbf{p}_{hand}(t)}{\|\mathbf{p}_i(t) - \mathbf{p}_{hand}(t)\|}. \quad (1.1)$$

These vectors geometrically define the directions in which the user is focusing attention or exerting effort. To incorporate dynamics of motion, normalized gaze, head orientation, and hand acceleration vectors are introduced as  $\hat{\mathbf{v}}_{gaze}(t)$ ,  $\hat{\mathbf{v}}_{head}(t)$ , and  $\hat{\mathbf{a}}_{hand}(t)$ .

**Cosine Similarity Scores** : To evaluate whether the user’s body movements align with the direction of candidate objects, cosine similarity between these features and the direction vectors is computed:

$$s_{gaze}(i, t) = \max(0, \hat{\mathbf{v}}_{gaze}(t) \cdot \mathbf{d}_i^{eye}(t)), \quad (1.2)$$

$$s_{head}(i, t) = \max\left(0, \hat{\mathbf{v}}_{head}(t) \cdot \mathbf{d}_i^{head}(t)\right), \quad (1.3)$$

$$s_{accel}(i, t) = \max\left(0, \hat{\mathbf{a}}_{hand}(t) \cdot \mathbf{d}_i^{hand}(t)\right). \quad (1.4)$$

Here,  $s_{gaze}$  represents the degree to which the gaze is directed toward each object,  $s_{head}$  represents the degree to which the head orientation captures each object, and  $s_{accel}$  represents the degree to which the hand acceleration is aligned with the object’s direction.

In addition, physical graspability is considered by introducing: (i) a proximity score  $s_{prox}$  based on the Euclidean distance between the hand and the object, and (ii) a binary

contact variable  $c_i(t)$ :

$$s_{\text{prox}}(i, t) = \text{clip} \left( 1 - \frac{\|\mathbf{p}_{\text{hand}}(t) - \mathbf{p}_i(t)\|}{R_{\text{max}}}, 0, 1 \right), \quad c_i(t) \in \{0, 1\}. \quad (1.5)$$

Thus, the module integrates both body movement and physical graspability. Here,  $c_i(t)$  is treated as a mandatory flag for the final decision and is not included in the feature vector.

**Score Integration** : The feature vector is defined as

$$\mathbf{s}(i, t) = [s_{\text{prox}}, s_{\text{gaze}}, s_{\text{head}}, s_{\text{accel}}]^\top, \quad (1.6)$$

and the integrated score is calculated with different weightings depending on whether the target belongs to the reachable set  $\mathcal{B}_{\text{in}}(t)$  or the unreachable set  $\mathcal{B}_{\text{out}}(t)$ :

$$S(i, t) = \begin{cases} \mathbf{w}_{\text{in}}^\top \mathbf{s}(i, t), & i \in \mathcal{B}_{\text{in}}(t), \\ \mathbf{w}_{\text{out}}^\top \mathbf{s}(i, t), & i \in \mathcal{B}_{\text{out}}(t). \end{cases} \quad (1.7)$$

Here,  $\mathbf{w}_{\text{in}}$  and  $\mathbf{w}_{\text{out}}$  are predefined weight vectors. For reachable objects, emphasis is placed on contact and proximity, ensuring robustness against noisy gaze fluctuations. For unreachable objects, emphasis is placed on gaze and hand motion, capturing the user's distal intention. This design enables flexible and error-tolerant inference depending on the situation.

**Temporal Smoothing** :

Since the score is expected to fluctuate rapidly due to momentary body movements, an exponential moving average is introduced to temporally smooth the score [18].

$$\bar{S}(i, t) = \alpha S(i, t) + (1 - \alpha) \bar{S}(i, t - 1), \quad \alpha \in (0, 1]. \quad (1.8)$$

Here,  $\bar{S}(i, t)$  is the smoothed integrated score, and  $\alpha$  is the smoothing coefficient. This prevents transient misdetections and allows stable inference of user intention.

**Final Decision Rule** : The final target is determined with priority given to contact. If there exists any candidate  $i$  such that  $c_i(t) = 1$ , that object is immediately selected:

$$i^*(t) \in \{i \in \mathcal{B} \mid c_i(t) = 1\}. \quad (1.9)$$

Otherwise, the target is selected as the one maximizing the smoothed score over the entire candidate set  $\mathcal{B}$ :

$$i^*(t) = \arg \max_{i \in \mathcal{B}} \bar{S}(i, t). \quad (1.10)$$

However, the selected object must satisfy threshold conditions:

$$\bar{S}(i^*(t), t) \geq \begin{cases} T_{\text{in}}, & i^*(t) \in \mathcal{B}_{\text{in}}(t), \\ T_{\text{out}}, & i^*(t) \in \mathcal{B}_{\text{out}}(t). \end{cases} \quad (1.11)$$

Here,  $T_{\text{in}}$  and  $T_{\text{out}}$  are thresholds for the reachable and unreachable sets, respectively. This two-step decision ensures that only targets with sufficiently strong intentional signals are confirmed.

**Information Presentation Adjustment** : When presenting inference results to the user, the TBI module adjusts the amount of information to prevent cognitive overload while supporting intuitive understanding.

First, as shown in Figure 1.4, each candidate bottle is classified by the *Reachability Map (RM)* as “graspable” or “non-graspable.” Graspable objects are displayed with a base color [17], while non-graspable ones are rendered semi-transparent with reduced saturation, ensuring they do not unnecessarily attract the user’s attention.

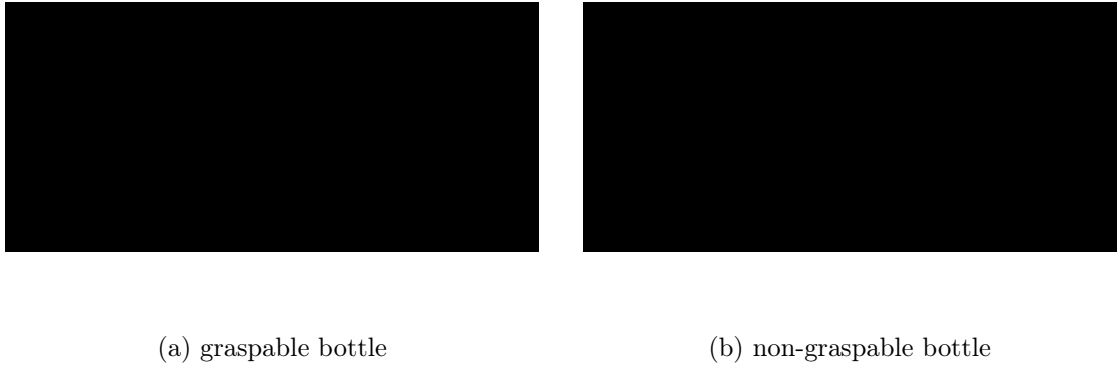


Fig.1.4: candidate bottle is classified by the Reachability Map

Furthermore, as illustrated in Figure 1.5, once the target is inferred, all non-selected objects are displayed in semi-transparent form. At this stage, different highlighting schemes are used for graspable and non-graspable objects: graspable targets are emphasized with a color indicating manipulability, while non-graspable targets are highlighted with a different tone to indicate the need for base relocation.

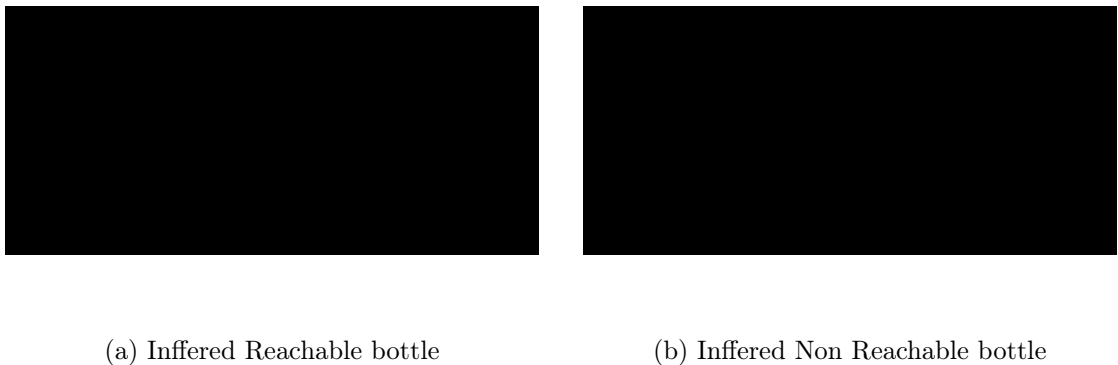


Fig.1.5: Inferred target bottle

Through this two-step information adjustment, users can clearly distinguish between “objects that can currently be grasped” and “objects that require relocation.” By emphasizing the final inferred target, the system reduces selection errors and alleviates cognitive load, supporting intuitive decision-making.

**Summary of TBI** : In summary, the TBI module geometrically defines multiple bodily features such as gaze, head orientation, and hand motion, and integrates them through

cosine similarity, proximity, and contact indicators to construct a robust scoring framework. Temporal smoothing suppresses transient misdetections, ensuring stable real-time inference.

Furthermore, the inference results are visually presented in the MR space. Specifically, depending on the Reachability Map (RM) judgment of whether an object is graspable or not, graspable objects are displayed in their base color, whereas ungraspable ones are shown with reduced information through transparency or lowered saturation. In addition, the bottle inferred as the operation target is highlighted, and different highlight colors are used for graspable and ungraspable objects, thereby providing clear feedback to the user.

In this way, the TBI module integrates both “target inference” and “information presentation,” reducing the user’s cognitive load while preventing misselection of operation targets. As a result, it enables the accurate understanding of human intention, which serves as the foundation of the entire HRI-CAS framework.

#### 1.4.2 Base Relocation based on Inverse Reachability Map (BRI)

When the target lies outside the robot’s reachable workspace, the robot base itself must be relocated. Naively exploring base candidates leads to prohibitive computational cost and makes it difficult to obtain configurations that respect human–robot collaboration. Therefore, as illustrated in Figure 1.6, we introduce two concepts widely used in manipulator research—the *Reachability Map (RM)* and the *Inverse Reachability Map (IRM)*—to generate and evaluate candidate base poses [17].

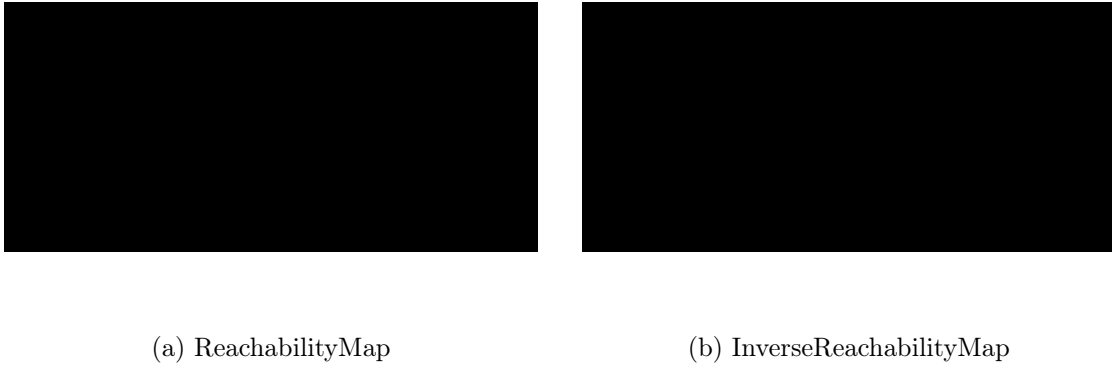


Fig.1.6: RM/IRM

#### Basics of reachability and inverse reachability :

Given a robot base pose  $\mathbf{b}$ , the workspace that the end-effector (EE) can reach from  $\mathbf{b}$  is defined as

$$RM(\mathbf{b}) = \{\mathbf{p} \in \mathbb{R}^3 \mid \exists \theta \text{ s.t. } FK(\theta, \mathbf{b}) = \mathbf{p}\}, \quad (1.12)$$

where  $FK(\theta, \mathbf{b})$  denotes forward kinematics. In other words, if there exists a joint configuration  $\theta$ , the point  $\mathbf{p}$  is reachable from the fixed base  $\mathbf{b}$ . Thus, the RM specifies

“how far the arm can reach from a fixed base pose.”

Conversely, for a given task point  $\mathbf{p}_t$ , the set of base poses that makes  $\mathbf{p}_t$  reachable is

$$\mathcal{B}(\mathbf{p}_t) = \{\mathbf{b} \in \mathbb{R}^3 \mid \mathbf{p}_t \in RM(\mathbf{b})\}, \quad (1.13)$$

which we call the *Inverse Reachability Map (IRM)*. IRM is the inverse mapping that provides “the set of base poses from which the task point is reachable,” and is essential when base relocation is considered [17].

Beyond a binary reachable/unreachable decision, RM can evaluate stability/redundancy by counting the number of inverse kinematic (IK) solutions available at a point. In this work, the RM score for position  $\mathbf{p}$  is defined by

$$S_{RM}(\mathbf{p}) = N_{IK}(\mathbf{p}). \quad (1.14)$$

A larger number of solutions implies greater redundancy and flexibility in motion, which typically increases grasp success rates.

Since IRM is the inverse of RM, each candidate base pose  $\mathbf{b} \in \mathcal{B}(\mathbf{p}_t)$  for target  $\mathbf{p}_t$  inherits the corresponding RM score:

$$S_{IRM}(\mathbf{b}; \mathbf{p}_t) = N_{IK}(\mathbf{p}_t \mid \mathbf{b}). \quad (1.15)$$

This allows us to assess candidate base configurations not only by “reachability,” but also by “how redundantly and stably” the target can be reached.

We now select the optimal base pose from the candidate set  $\mathcal{B}(\mathbf{p}_t)$ . As shown conceptually in Figure 1.7, we formulate two cases depending on the availability of environment/navigation information.

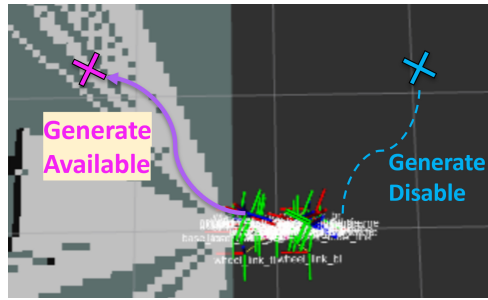


Fig.1.7: Figure7

### Case: Known path

When a navigable path to the target is available, the base pose is selected by jointly considering the IRM score  $S_{IRM}(\mathbf{b}; \mathbf{p}_t)$  and the movement cost  $C(\mathbf{b})$ :

$$\mathbf{b}^* = \arg \max_{\mathbf{b} \in \mathcal{B}(\mathbf{p}_t)} [S_{IRM}(\mathbf{b}; \mathbf{p}_t) - \lambda C(\mathbf{b})]. \quad (1.16)$$

Here,  $C(\mathbf{b})$  is defined as the navigation path length from the current base  $\mathbf{b}_0$  to  $\mathbf{b}$ :

$$C(\mathbf{b}) = L(\Gamma(\mathbf{b})) = \int_0^1 \|\dot{\mathbf{x}}(s)\| ds \approx \sum_{k=1}^N \|\mathbf{x}_k - \mathbf{x}_{k-1}\|.$$

The resulting candidate base poses and the waypoint sequence are visualized in MR space, and the user can modify the route by directly manipulating the waypoints.

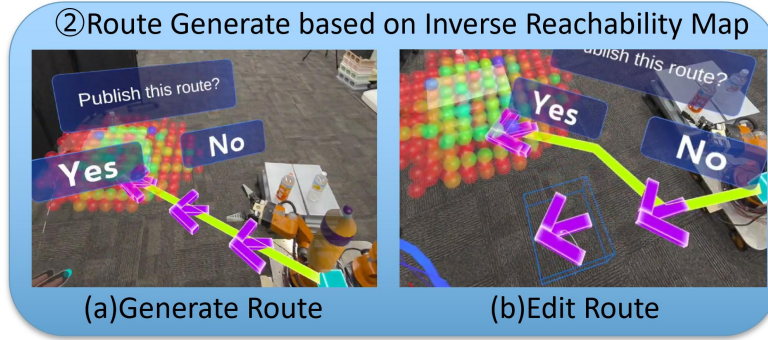


Fig.1.8: Figure4

**User interaction (mathematical incorporation of path edits)** Let the reference route edited by the user in MR space be  $\bar{\Gamma}_u = \{\bar{\mathbf{x}}_k\}_{k=0}^N$ , and let  $\mathcal{K}_{\text{lock}}$  denote the set of pinned (fixed) indices. For each candidate base  $\mathbf{b}$ , the navigation module recomputes a path  $\Gamma(\mathbf{b})$  by minimizing

$$C(\mathbf{b} \mid \bar{\Gamma}_u) = C(\mathbf{b}) + w_{\text{trk}}^{\text{nav}} \sum_{k \notin \mathcal{K}_{\text{lock}}} \|\mathbf{x}_k - \bar{\mathbf{x}}_k\|^2, \quad \text{s.t. } \Gamma(\mathbf{b}) \in \mathcal{F}_{\text{nav}}, \quad \mathbf{x}_k = \bar{\mathbf{x}}_k \quad (k \in \mathcal{K}_{\text{lock}}).$$

Then (1.16) is re-evaluated using the updated  $\Gamma(\mathbf{b})$ . The final base is chosen as

$$\mathbf{b}^* = \arg \max_{\mathbf{b} \in \mathcal{B}(\mathbf{p}_t) \cap \mathcal{F}_{\text{nav}}} \left[ S_{\text{IRM}}(\mathbf{b}; \mathbf{p}_t) - \lambda C(\mathbf{b} \mid \bar{\Gamma}_u) - \xi \Delta \psi(\mathbf{b})^2 \right],$$

and the updated waypoint sequence  $\Gamma(\mathbf{b}^*)$  is immediately re-rendered in MR space.

### Case: Unknown path

When environment information is incomplete and path generation fails, we adopt the IRT-MRO [14] scheme: the user directly lays out a base route in MR space. On top of this, we introduce a local autonomous refinement using a cost map. As shown in Figure1.9, each waypoint is adjusted to a safe and feasible pose that accounts for obstacles and environmental changes. The refined route is immediately reflected in MR space so that the user can visually grasp the autonomous corrections made by the robot.

**Cost map (OccupancyGrid  $\rightarrow$  safety-aware grid) :**

From an occupancy grid  $G \in \{-1, 0, 100\}^{H \times W}$  (unknown = -1, free = 0, obstacle = 100), we construct a cost map  $C \in \mathbb{R}^{H \times W}$ . The per-cell base cost is

$$C_{ij} = \begin{cases} 100 & (G_{ij} = 100) \quad [\text{blocked}] \\ P_{\text{unk}} & (G_{ij} = -1) \quad [\text{mild penalty for unknown}] \\ 0 & (G_{ij} = 0), \end{cases}$$

followed by two dilations for safety margins. (i) Obstacle-proximity dilation by Manhattan distance  $d_1 \leq d_{\text{avoid}}$  sets  $C_{ij} \leftarrow P_{\text{pen}}$  (morphological expansion). (ii) Robot-radius expansion: with resolution  $h$  and safety radius  $r_{\text{safe}}$ , we set  $r_c = \lceil r_{\text{safe}}/h \rceil$  and also assign  $C_{ij} \leftarrow P_{\text{pen}}$  for all cells within Euclidean distance  $\leq r_c$  of any obstacle. Here  $P_{\text{pen}} \gg 1$  encodes “traversable but expensive,” thus naturally favoring paths that keep distance from obstacles (cf. Fig. 1.9).

**Route proposal (MR reference  $\rightarrow$  safe A\* refinement) :**

Let the user-provided MR reference route be  $\bar{\Gamma} = \{\bar{\mathbf{x}}_k\}_{k=0}^N$ , treated as a sequence of waypoints. For each segment ( $\bar{\mathbf{x}}_{k-1} \rightarrow \bar{\mathbf{x}}_k$ ), we run A\* on the safety-aware cost map to obtain a safe path  $\Pi_k$ . We use 4-neighbor moves with per-step transition cost

$$\ell(i \rightarrow j) = 1 + \mathbb{I}\{C_j = P_{\text{pen}}\} P_{\text{pen}}, \quad h(n) = |x_n - x_g| + |y_n - y_g| \quad (\text{Manhattan heuristic}),$$

and forbid transitions through blocked cells ( $C = 100$ ). The heuristic  $h$  is admissible and consistent, so A\* returns an optimal path under  $\ell$ . In practice, we also compute a variant without penalties  $\Pi_k^{\text{nop}}$  and pick, for each segment, the path with the lower total cost; the final route is  $\Gamma = \Pi_1 \oplus \dots \oplus \Pi_N$ . For execution, we resample  $\Gamma$  every  $\delta$  meters to produce a concise waypoint list and follow it under velocity saturation in the robot frame.

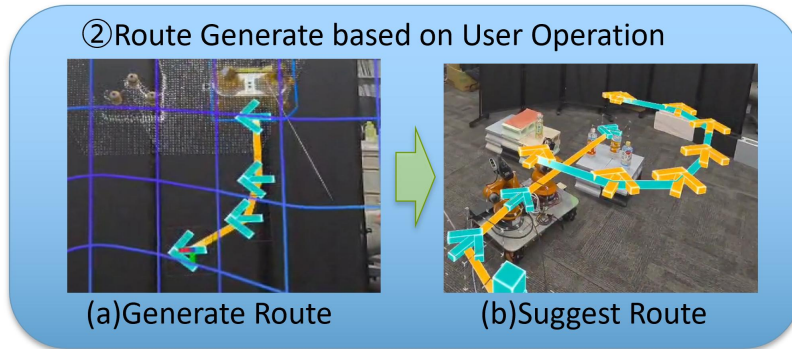


Fig.1.9: Figure4



## Summary of BRI

In summary, our base relocation strategy adopts a two-track structure: (i) when a path is known, we perform autonomous optimization based on the IRM score and movement cost; (ii) when a path is unknown, we combine user-specified routes with cost-map-based refinement. This design satisfies both robot autonomy and user-driven interaction, achieving flexible and robust base relocation.

### 1.4.3 Human-Intent Adaptive Grasp Manipulation (HIGM)

Even when the base position is appropriate, discrepancies between human intent and the robot's kinematic constraints can cause the End-Effector (EE) to approach the target in unstable poses. This can lead to grasp failure or excessive joint load. To address this issue, we propose an *adaptive grasp framework* that integrates Pregrasp design, null-space optimization, and a finite state machine (FSM).

**Pregrasp and EE interpolation:** Human operation alone may command poses that are difficult for the robot to execute. For example, if the user attempts to pinch the top of a bottle, directly mapping the hand-back pose to the EE could result in an unstable grasp configuration.

To address this, as shown in Figure 1.10, we generate a robot-friendly reference pose (Pregrasp pose  $\mathbf{x}_{pre}$ ) based on the target position  $\mathbf{p}_t$ . The actual EE target is defined by interpolating between the user's palm pose  $EE_{palm}$  and the Pregrasp pose  $EE_{pre}$ , depending on the distance  $d$  to the target:

$$EE_{target}(d) = (1 - \beta(d)) EE_{palm} + \beta(d) EE_{pre}, \quad (1.17)$$

where  $\beta(d) \in [0, 1]$  is a distance-dependent interpolation coefficient.

With this scheme, user intent ( $EE_{palm}$ ) dominates when the target is far away, and as the robot approaches, the weight gradually shifts toward the stable Pregrasp pose ( $EE_{pre}$ ). In other words, the user's motion is “magnetically guided” toward the object—similar to aim assist in video games—preserving intuitive control while ensuring stable and safe grasping for the robot.

Furthermore, Pregrasp interpolation provides the global reference, while null-space optimization ensures local stability and safety. Thus, interpolation defines “which pose should be aimed for,” and null-space optimization ensures “how to safely approach that pose.” Combining them achieves grasping that respects user intent while adapting to the robot's kinematic limits.

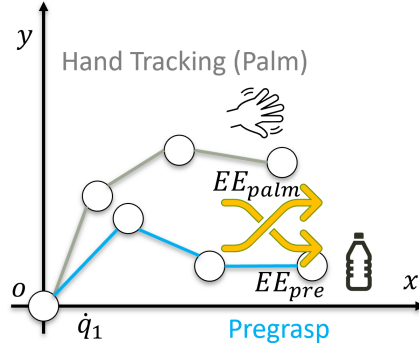


Fig.1.10: Figure10

**Joint refinement through null-space optimization:** For redundant manipulators, multiple joint configurations can realize the same EE pose. The set of joint motions that do not affect the EE is called the *null space* [19][20].

Formally, the task-space velocity  $\dot{x}$  and joint velocity  $\dot{\theta}$  satisfy

$$\dot{x} = J\dot{\theta}. \quad (1.18)$$

A velocity  $\dot{\theta}_{null}$  belongs to the null space if

$$J\dot{\theta}_{null} = 0. \quad (1.19)$$

Such motions leave the EE unaffected while adjusting only the joint posture.

We exploit this property to simultaneously achieve the primary task and optimize joint posture. The joint velocity is given by

$$\dot{\theta} = J^+ \dot{x}_d + P \nabla_{\theta} H(\theta), \quad (1.20)$$

where the damped least-squares pseudoinverse is

$$J^+ = J^{\top} (J J^{\top} + \lambda I)^{-1}, \quad (1.21)$$

and  $P = I - J^+ J$  is the null-space projector. The first term guarantees EE tracking, faithfully reproducing the user's hand motion. The second term optimizes posture in the null space using the merit function  $H(\theta)$  (Figure1.10):

$$H(\theta) = w_{mani} \log \det(J J^{\top}) + w_{limit} \sum_i \log m_i(\theta) - \gamma \|\theta - \theta_{pre}\|^2. \quad (1.22)$$

The roles of each term are:

- $w_{mani} \log \det(J J^{\top})$ : maximizes Yoshikawa's manipulability[21], avoiding singularities and preventing the arm from losing freedom. This maintains motion redundancy, providing the user with smooth and continuous operation.
- $w_{limit} \sum_i \log m_i(\theta)$ : enlarges the margin  $m_i(\theta)$  to joint limits[22], preventing joints from approaching boundaries. This ensures mechanical safety and reduces the risk of sudden motion stops, improving user comfort.

- $-\gamma\|\theta - \theta_{pre}\|^2$ : maintains consistency with the Pregrasp pose, preventing optimization from drifting too far from user intent. This balances intuitive operation and kinematic stability.

Thus, null-space optimization ensures (1) stability through singularity avoidance, (2) safety through joint-limit avoidance, and (3) intent preservation through Pregrasp consistency. It functions not only as numerical stabilization, but as a foundation linking “robotic physical constraints” with “the user’s continuous intuitive experience.”

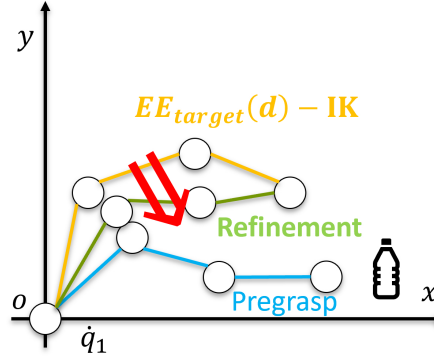


Fig.1.11: Figure11

**Grasp Readiness Finite State Machine:** To ensure that the entire grasping operation proceeds safely and stably, this study introduces a Finite State Machine (FSM) consisting of four states. A Finite State Machine (FSM) is a classical modeling approach that defines system behavior as a combination of states and state transitions [23] [24]. Each state specifies the system’s behavior, and transitions to other states occur in response to external inputs or internal conditions. This simple yet explicit framework makes it possible to manage complex actions step by step, thereby facilitating safety and stability assurance. Through the FSM, both the reflection of user intent and the accommodation of the robot’s kinematic constraints are guaranteed, preventing failures such as the robot “approaching the target in an unstable posture.”

The roles and transitions of the states are:

1. **TRACK:** The robot follows the target using Palm/Pregrasp interpolation. Mathematically, the EE target pose  $EE_{target}(d)$  is generated and joint angles are updated via DLS-IK. User experience: “the robot hand smoothly follows my hand.”
2. **BACKOFF:** If manipulability  $M(\theta) < \tau_{mani}$  or margin  $m_i(\theta) < \tau_{limit}$ , the robot retreats to avoid instability. This prevents “robot jamming” and provides the user with a sense of safety.
3. **REALIGN:** After BACKOFF, null-space optimization realigns the joints until  $M(\theta) \geq \tau_{mani}$  and  $m_i(\theta) \geq \tau_{limit}$ . User experience: “the robot autonomously reconfigures itself,” revealing its adaptiveness.

4. **APPROACH:** Once stability is secured and the target distance  $d < \tau_{dist}$ , the EE approaches to grasp. This corresponds to “the robot decisively grabs,” improving grasp success.

The FSM links mathematical criteria (manipulability, joint margin, distance thresholds) with user experience (tracking feel, safety, adaptiveness, visible grasp action). In particular, the BACKOFF  $\rightarrow$  REALIGN transition guarantees that even if the user makes errors, the robot autonomously corrects them, realizing “collaborative safety” in HRC.

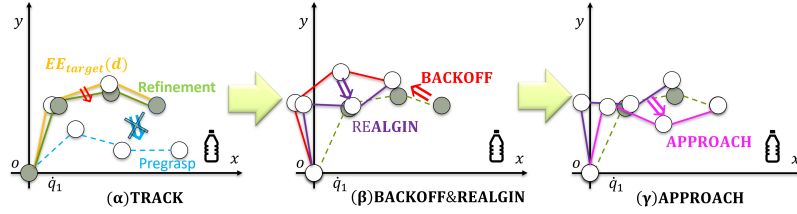


Fig.1.12: Figure12

#### 1.4.4 Significance of Module Integration

The proposed HRI-CAS consists of three modules: Target Bottle Inference (TBI), Base Relocation based on IRM (BRI), and Human-Intent Adaptive Grasp Manipulation (HIGM). Each module functions independently, but more importantly, they complement each other to overcome the limitations faced by conventional MR-based HRC systems.

First, TBI accurately estimates the user’s intent, clarifying “which object should be manipulated.” This object information is then passed to BRI, which determines whether base relocation is necessary depending on the object’s reachability. Subsequently, HIGM leverages the outputs of TBI and BRI to compensate for discrepancies between human intent and robot kinematic constraints, ensuring stable grasping.

Through the integration of TBI, BRI, and HIGM, the system establishes a coherent workflow: (1) recognizing user intent with high precision, (2) ensuring reachability through appropriate base relocation, and (3) guaranteeing grasp stability. As a result, the robot operates intuitively and adaptively without over-relying on the user’s physicality, providing a stable interaction experience for both novice and expert users.

Thus, the significance of this research lies not merely in individual technical improvements, but in the establishment of an integrated framework for HRC in MR environments,

encompassing the flow of “intent recognition  $\rightarrow$  reachability assurance  $\rightarrow$  stable grasp.”

## 1.5 Experiment on BRI

### 1.5.1 Experimental Setup

We fixed the BRI evaluation to the *known-path* case (navigation available from the occupancy grid and A\*). The user-edited route (unknown-path) variant is reported as an ablation in the appendix.

To verify the effectiveness of the proposed base relocation strategy based on the Inverse Reachability Map (BRI), a stepwise pick-and-place experiment was designed. The experiments were conducted using the KUKA YouBot platform, which integrates omni-directional base mobility with a 5-DOF manipulator. Plastic bottles were used as target objects, and they were deliberately placed at positions that were unreachable from the robot’s initial base position.

The user was positioned approximately 1.0 m away from the robot and monitored and assisted the task through an MR device.

The experiments were divided into two phases. In **Phase 1**, the robot base was relocated once using the BRI strategy, followed by a single pick-and-place operation of one bottle. The objective of this phase was to confirm whether the robot could move to a position where grasping was physically feasible, in accordance with the optimal solution computed from the IRM score and relocation cost. In **Phase 2**, an additional base relocation was performed, after which the robot was tasked with sequentially executing pick-and-place operations for two bottles. This phase aimed to examine whether BRI could maintain task stability through multiple relocations. For the BRI evaluation, a single participant ( $n=1$ ) performed 10 repeated trials.

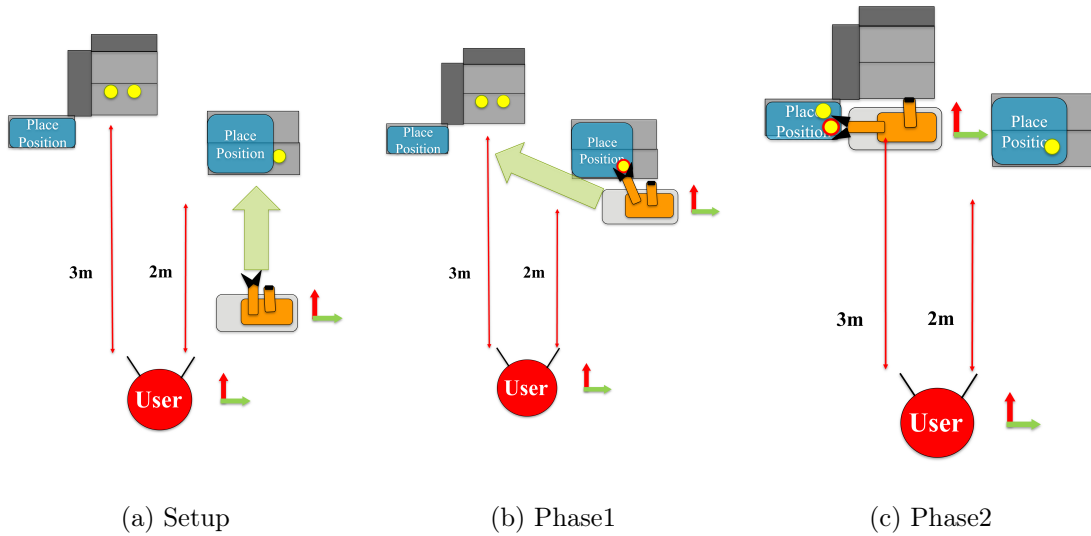


Fig.1.13: Overview of BRI pipeline

The evaluation metrics were as follows: (i) grasp success rate, defined as the percentage of trials in which the target object was correctly picked up and placed at the designated

position; (ii) accuracy of the relocation decision, i.e., whether the base positions computed by BRI corresponded to physically feasible grasping positions; and (iii) time required for base relocation as well as the overall task completion time.

We define the “optimal-base hit rate” as the proportion of trials in which the final executed base pose  $\mathbf{b}^*$  yields at least one feasible IK solution at the task point:

$$\text{Hit} \Leftrightarrow N_{\text{IK}}(\mathbf{p}_t \mid \mathbf{b}^*) > 0$$

OccupancyGrid values were mapped to the cost map with obstacle= 100, unknown= 30, free= 0, and a safety dilation penalty  $P_{\text{pen}} = 50$ . Robot radius for obstacle expansion was fixed to 0.4 m. Waypoint resampling interval was 0.5 m. Path search used 4-neighbor A\* with Manhattan heuristic and per-step cost 1 (free) or  $1+P_{\text{pen}}$  (penalized cells). When both penalized and non-penalized routes existed, the lower-cost path was selected. All parameters were fixed across participants.

Through this experimental design, the aim was to demonstrate that BRI can autonomously and efficiently relocate the robot base to enhance both grasp success rate and task stability, not only for single-object manipulation but also for tasks involving multiple objects.

This single-subject setup was adopted as a pilot evaluation to verify feasibility and repeatability of the BRI pipeline.

## 1.5.2 Experimental Results

# 1.6 Experiment2

## 1.6.1 Experimental Setup

To clarify the advantages of the proposed Human-Intent Adaptive Grasp Manipulation (HIGM), we conducted a comparative evaluation against the existing MR-based manipulation framework, Immersive Robot Teleoperation Based on User Gestures in Mixed Reality Space (IRT-MRO) [14]. The conventional system places emphasis on directly transferring user operations in MR space to the robot side, without sufficiently considering kinematic feasibility or grasp stability.

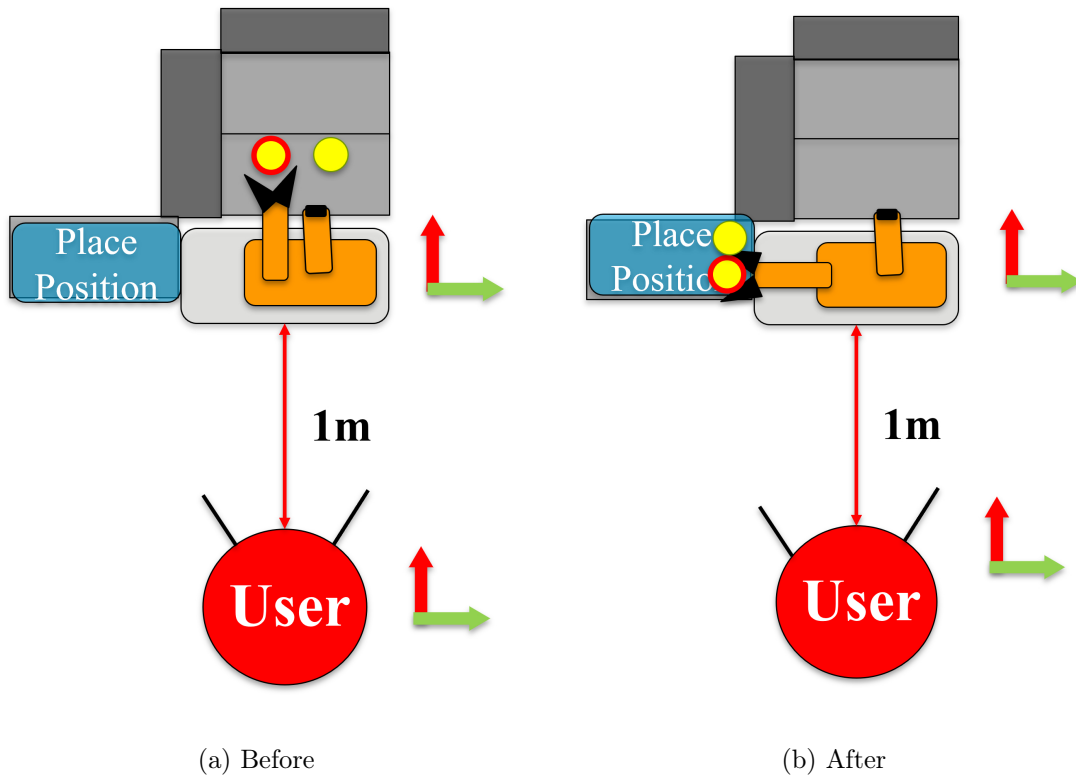
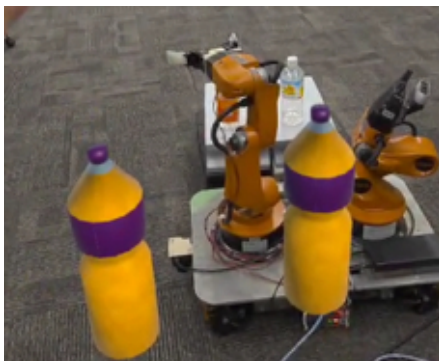


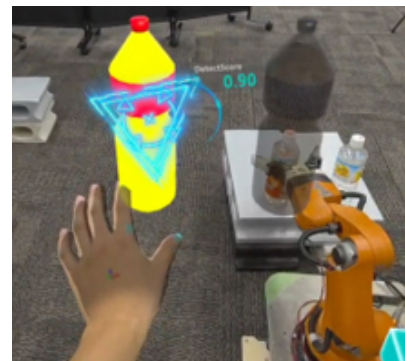
Fig.1.14: Sequential pick-and-place before/after

The experimental setup was designed to evaluate both the baseline performance of IRT-MRO and the improvements introduced by HIGM. Figure 1.14 illustrates an example of a pick-and-place experiment using the proposed system. Using the KUKA YouBot platform, the user was positioned 1.0 m away from the robot, and the robot was assigned the task of placing two bottles, located within its graspable range, into a designated box. The task was considered successful if both bottles were correctly placed inside the box. In this study, it was assumed that the user was able to perceive the task environment.

As shown in Figures 1.15 a and b, when the user grasped the MR object, the robot simultaneously grasped the corresponding real object. Similarly, when the user released the MR object, the robot released the real object as well (Figures 1.16c and d). Through this sequence of procedures, the pick-and-place task was executed.



(a) before



(b) after

Fig.1.15: Sequential pick-and-place



(a) before

(b) after

Fig.1.16: Sequential pick-and-place

The evaluation was conducted in terms of task execution time measured from initiation to completion, workload assessed using the NASA-TLX[25], the degree of convergence toward the pregrasp posture (adhesion), the frequency of singularities and joint-limit violations, and the number of regrasp attempts. We fixed all thresholds and weights as follows.

$$\beta(d) = \text{hyst-sigmoid}(d; d_{\text{in}}=0.18 \text{ m}, d_{\text{out}}=0.22 \text{ m}, k=40, \alpha=0.2),$$

DLS damping  $\lambda = 10^{-2}$  (adaptive around  $e + 0.002$  in implementation), FSM thresholds  $\tau_{\text{mani}}$  (min. singular value) = 0.03,  $\tau_{\text{cond}}$  (condition number) = 60,  $\tau_{\text{limit}}$  (min. joint margin) = 0.08 rad. Backoff distance 0.07 m, REALIGN timeout 0.6 s, and null-space step  $k_{\text{null}} = 0.02$ . The merit weights are  $w_{\text{mani}} = 1.0$ ,  $w_{\text{limit}} = 0.3$ , and a weak pull to pregrasp. These values are held constant across participants and trials.

Each participant performed the tasks under both systems: the baseline system (IRT-MRO) and the proposed system (HIGM). The order of conditions was counterbalanced across participants to avoid ordering bias. Each task was repeated 10 times per condition, resulting in 20 trials per participant. A total of 6 participants were recruited (n=6). All participants provided informed consent prior to the experiment, and the study protocol was approved in accordance with institutional guidelines.

We evaluate the Target Bottle Inference (TBI) *within* the HIGM user study rather than running a separate experiment, because HIGM can only act after TBI has decided the intended target. For each target-selection event, the ground truth  $g$  is the object that first receives user contact ( $c_i(t) = 1$ ); if no contact occurs, we use the object actually grasped and placed. TBI issues a decision at time  $t_{\text{dec}}$  when  $\max_i \bar{S}(i, t)$  exceeds the reachability-dependent threshold and is sustained for  $\tau_{\text{hold}}$ , and we set  $\hat{i} = \arg \max_i \bar{S}(i, t_{\text{dec}})$ . We report (i) top-1 accuracy  $\text{Acc}$ , also split by reachability ( $\text{Acc}_{\text{in}}/\text{Acc}_{\text{out}}$ ), (ii) the confirmation rate (fraction of events for which a decision is issued), and (iii) the post-decision switch rate (fraction of trials where the finally grasped object differs from  $\hat{i}$  due to later user changes). This protocol quantifies TBI under realistic collaboration while keeping the grasp-stability analysis attributable to HIGM.



Under these conditions, a comparative experiment was carried out to verify the performance of both systems.

### **1.6.2 Experimental Results**

## **1.7 Conclusions**

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